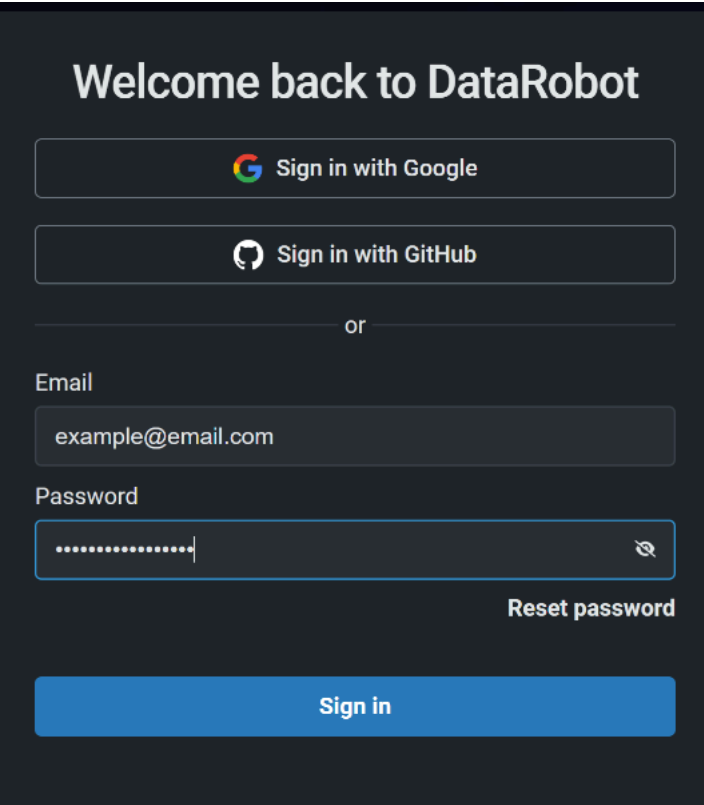
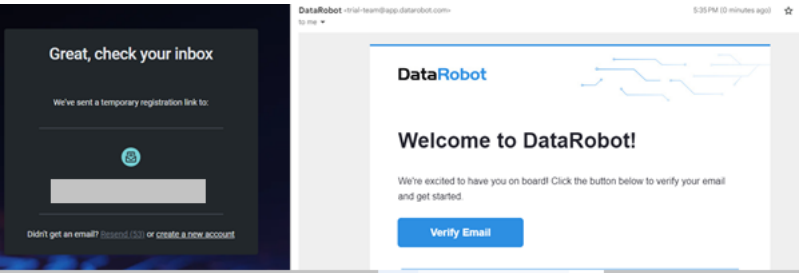
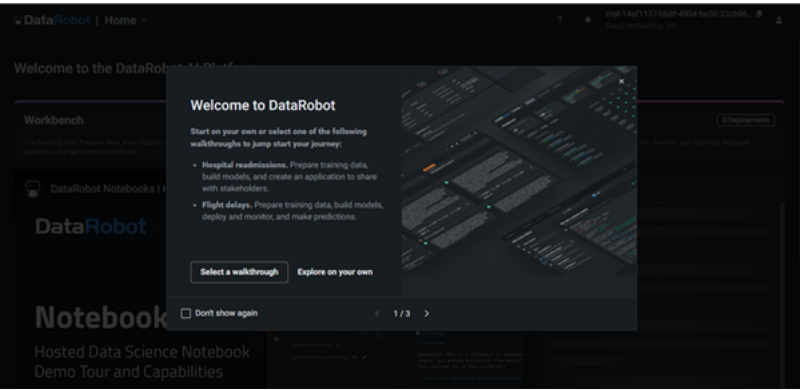




Step 4: Confirm your email by clicking **Verify Email** in your inbox.

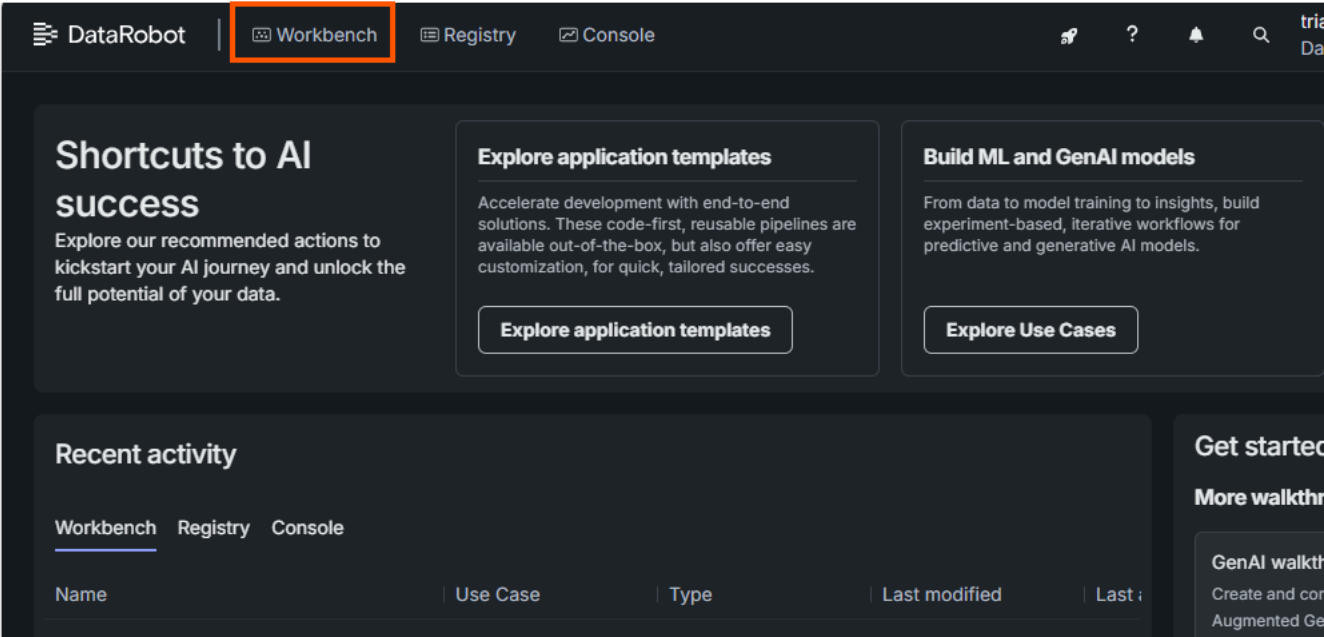


Step 5: Sign up and start your first experience of using the Generative AI tool. The dashboard will look like the image below. You may like to familiarize yourself with the application by clicking **Select a walkthrough**.

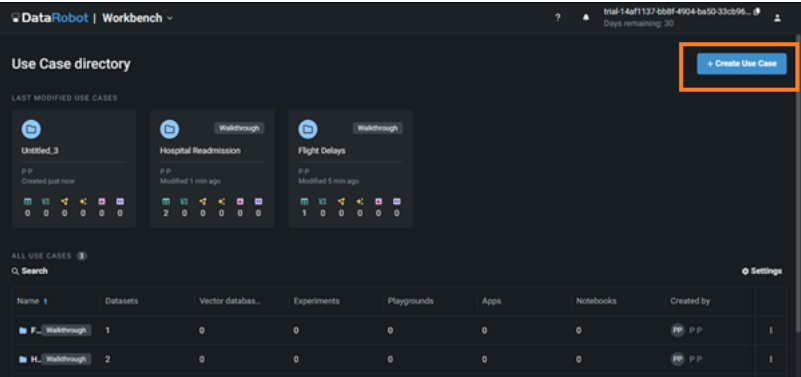


Task 2: Add a data set

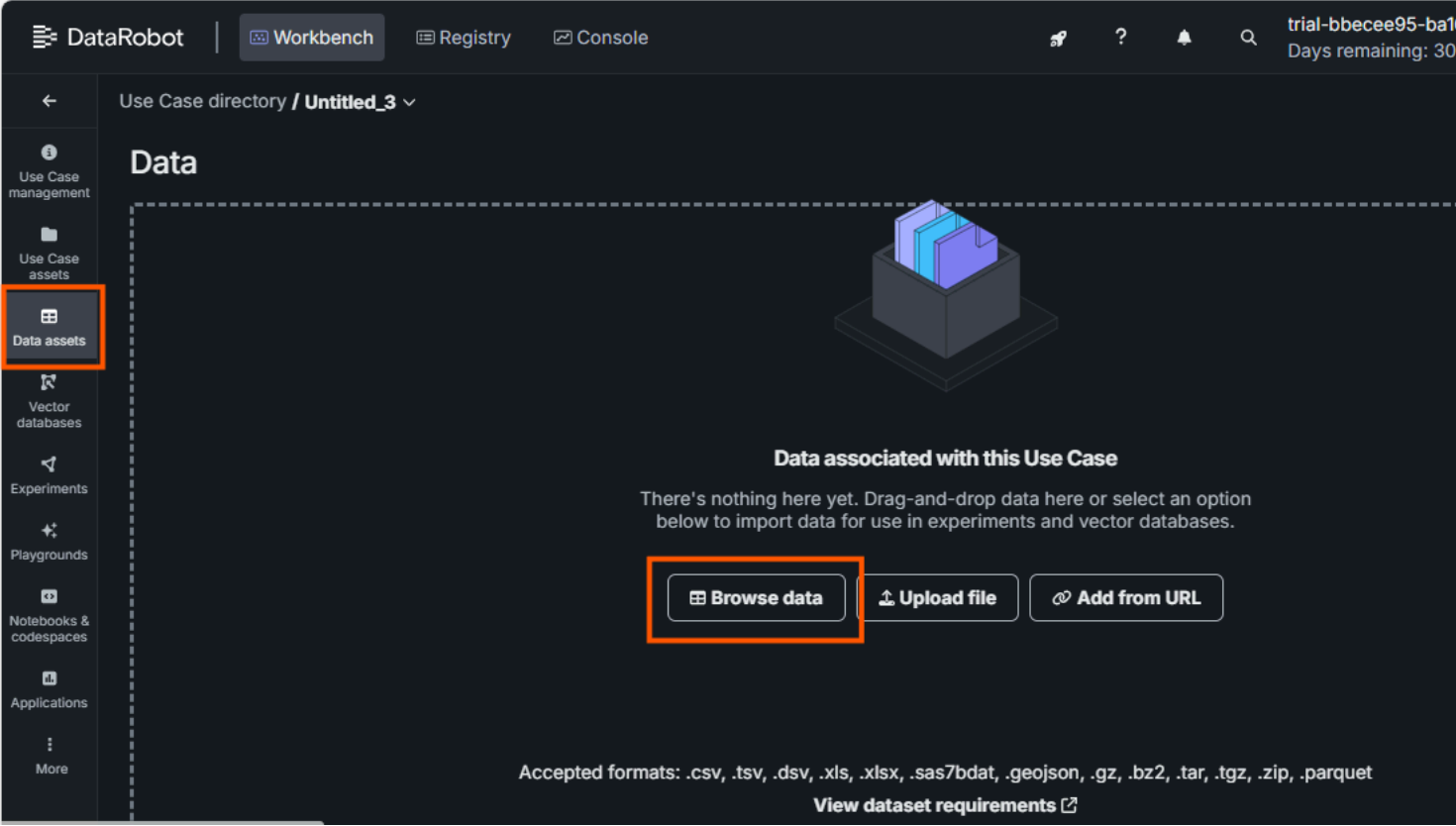
Step 8: The dashboard will appear shortly, and your screen will look as shown below. Click **Workbench**.



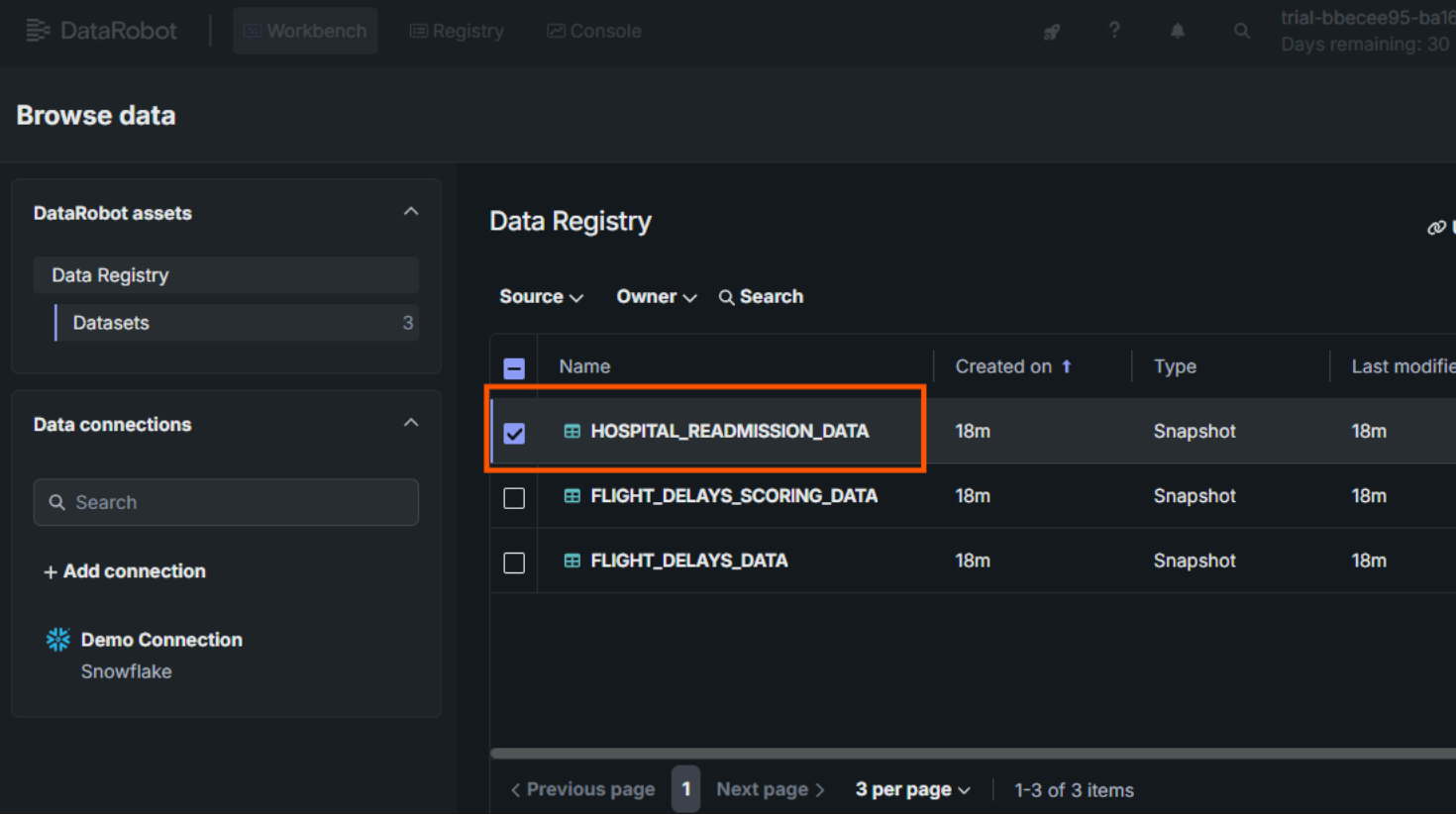
Step 9: Click Create Use Case.



Step 10: Click **Data assets** and select **Browse Data** to include the data set in your use case.



Step 10: You can select an in-built sample data set *HOSPITAL_READMISSION_DATA* and select **Add to Use Case**.



Step 11: Once you select the data set, you can see a preview of it by clicking on the dataset. You can also view the data set's features, as shown below. Click **Add to Use Case**.

DataRobot | Workbench

Workbench directory / Untitled_4

?

trial-c3398e81-5e8

Days remaining: 2

Add data

DATA STORES

↑ Upload

+ Connect

Q Search

Data Registry

DataRobot

Demo Connection

Snowflake

HOSPITAL_READMISSION_DATA

Data Registry / HOSPITAL_READMISSION_...

Features

Data preview

race	gender	age	weight	admission_type...
Categorical	Categorical	Categorical	Categorical	Categorical
Caucasian	Male	[50-60)		Emergency
Caucasian	Female	[70-80)		Urgent
Caucasian	Male	[60-70)		Elective
Caucasian	Male	[60-70)		
Caucasian	Female	[40-50)		Not Available
Caucasian	Male	[90-100)		Emergency
Caucasian	Male	[70-80)		Urgent
Caucasian	Female	[80-90)		Urgent
Missile	Female	[50-60)		Emergency

Snapshot sample

51 features | 1,511 rows

Step 12: After you add the data set to the use case, the workbench will appear as shown below. You can click the data set to see the feature insights.

DataRobot | Workbench

Workbench directory / Untitled_4

?

trial-c3398e81-5e8

Days remaining: 2

< Use Case directory

USE CASE

Untitled_4

:

Add Data

▼

:: All

Data

1

Vector databases

Experiments

Playgrounds

Applications

Type ▼

Q Search

Settings

Name	Created By	Last Modified ↑	Type	Source	Rows	
HOSPITAL_READMISSION_DATA	PV Pratiksha V.	now	Snapshot	Snowflake	10000	:

< Previous

1

Next >

1 per page ▼

1-1 of 1 items

i Use Case

DESCRIPTION

Add a de

CREATE

July 29,

LAST M

just now

OWNER

PV

Pr

TEAM

None

Mana

YOUR RO

Owner

Step 13: Explore the All Features menu to display specific features.

about:blank

5/10

DataRobot | Workbench Workbench directory / Untitled_4 / **HOSPITAL_READMISSION_DATA** Jul 29th, 2024 10:39 AM Snapshot Data actions

HOSPITAL_READMISSION_DATA

Data preview **Features** Feature lists

Show insights Show features from: All Features + Create feature list

Search

DATAROBOT FEATURE LISTS

- All Features 51
- Informative Features 40
- Raw Features 51

race age weight admission_type_id discharge

Caucasian 21% Male 54% [70-80] 26% ==Missing== 96% Emergency 49% Discharge

AfricanAmerican 6% 46% [60-70] 22% [75-100] 2% Urgent 19% Discharge

Other 52% Other 2% Other 32% Other

Caucasian Male [50-60]

Caucasian Female [70-80]

Caucasian Male [60-70]

Snapshot sample 51 features | 1,511 rows

Task 3: Work on Data Modeling

Step 14: You will have options **Modelling** and **Start wrangling**. You can try data wrangling if you want to. For this lab, you will work on model building under **Data actions** and select **Modelling**. It will take a while to prepare a data set for modelling.

DataRobot | Workbench Workbench directory / Untitled_4 / **HOSPITAL_READMISSION_DATA** Jul 29th, 2024 10:39 AM Snapshot Data actions

HOSPITAL_READMISSION_DATA

Data preview Features Feature lists

Show insights Show features from: All Features + Create feature list

Start wrangling

Start modeling

Start feature discovery

Download dataset

Remove dataset

race gender age weight admission_type_id discharge

Caucasian 74% Female 54% [70-80] 26% ==Missing== 96% Emergency 49% Discharge

AfricanAmerican 21% Male 46% [60-70] 22% [75-100] 2% Urgent 19% Discharge

Other 6% 52% Other 2% Other 32% Other

Caucasian Male [50-60]

Caucasian Female [70-80]

Caucasian Male [60-70]

Snapshot sample 51 features | 1,511 rows

Step 15: Once done, you need to select the **Target feature**. Select **readmitted** as your target feature.

DataRobot | Workbench

Workbench directory / Untitled_4

?

trial-c3398e81-5e

Days remaining: 2

Set up new experiment

Dataset

Target

Additional settings

Target feature

Select the feature to make predictions on.

payer_code

pioglitazone

race

readmitted

repaglinide

troglitazone

tolbutamide

tolazamide

time_in_hospital

			Uniq...	Missi...	Mean	Std Dev
			7	9592	-	-
troglitazone	35	Categorical	1	0	-	-
tolbutamide	30	Categorical	2	0	-	-
tolazamide	36	Categorical	2	0	-	-
time_in_hospital	7	Numeric	14	0	4.43	3.021

Experiment summary

HOSPITAL_READM

07-30 11:23:57

Dataset

Name

Rows

Features

Target

No target selected

Step 16: The workbench screen will be displayed as shown below. Click Next.

DataRobot | Workbench

Workbench directory / Untitled_4

?

trial-c3398e81-5e

Days remaining: 2

Set up new experiment

Dataset

Target

Additional settings

Target feature

Select the feature to make predictions on.

readmitted

Target type: Binary classification

Positive class: ☐ 0 ☒ 1

Modeling mode

Set the mode used for selecting which blueprints to build when training models.

Quick Autopilot

Optimization metric

Set the metric used when training models to evaluate and optimize accuracy.

LogLoss (Accuracy)

Recommended

Number of rows

False

True

Values of readmitted

Experiment summary

HOSPITAL_READM

07-30 11:23:57

Dataset

Name

Rows

Features

Target

Feature

Target type

Positive class

Modeling mode

Optimization metric

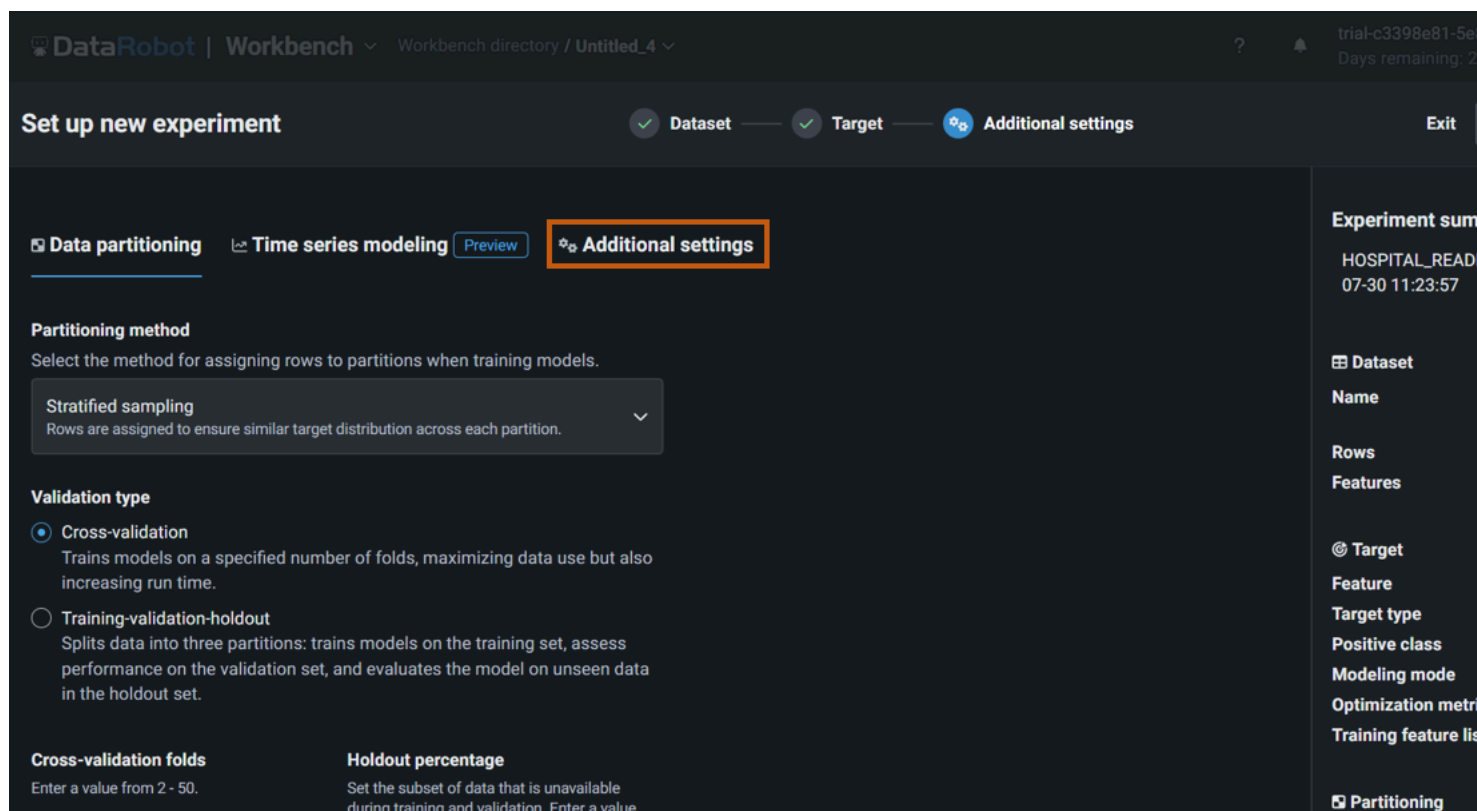
Training feature list

Partitioning

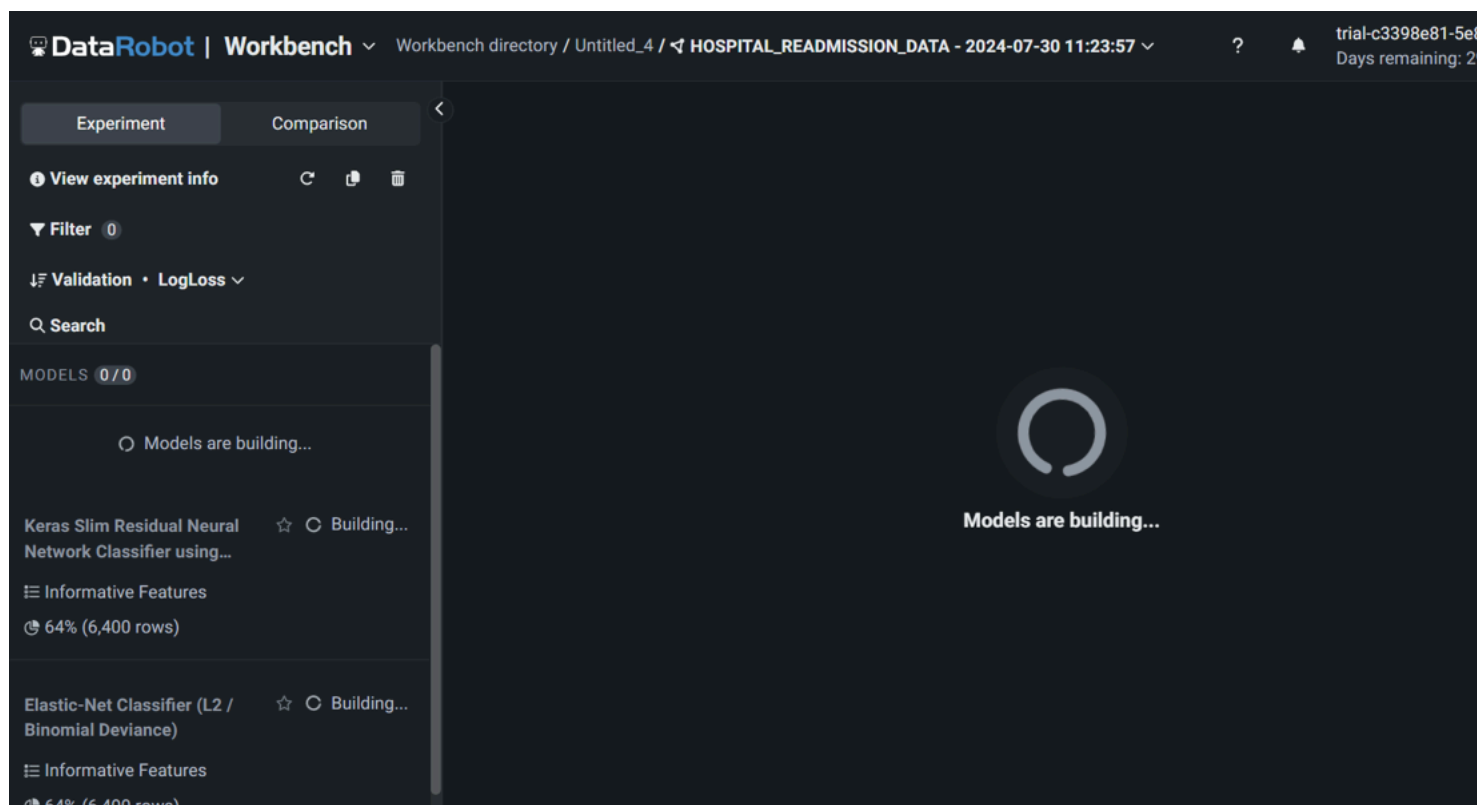
Step 17: You can modify the model setting in Additional Settings; once done, click Next and then click Start modelling.

about:blank

7/10



Step 18: Building models will take a while.



Step 19: once the modelling is complete, you can pick a model of your choice, and the DataRobot will show the **Model Overview**.

Experiment

Comparison

View experiment info

Filter

Validation • LogLoss

Search

MODELS 3/3

Models are building...

Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

Informative Features

64% (6,400 rows)

Elastic-Net Classifier (L2 / Binomial Deviance)

Informative Features

64% (6,400 rows)

Keras Slim Residual Neural

0.6327

Model Overview

Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

Training scores: LogLoss

Validation 0.6089

Cross-validation Score

Holdout 0.6111

Training settings

Training feature list

Training sample size

Informative Features 64% (6,400 rows)

Blueprint

Feature Impact

Feature Effects

Individual Prediction Explanations

ROC Curve

Step 20: You can explore various model overview components like **Blueprint**, **Feature Impact**, and so on.

Experiment

Comparison

View experim...

Filter

Validation • LogLoss

Search

MODELS 3/3

Models are building...

Generalized

Additive2...

Informative Features

64% (6,400 rows)

Light

Gradient...

Blueprint

Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

One-Hot Encoding | Missing Values Imputed | Standardize | Matrix of word-grams occurrences | Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

ElasticNet Generalized Linear Model

Click a blueprint node to view more information. View more blueprints in the [blueprint repository](#).

Search

Edit blueprint

Data

Categorical Variables

Numeric Variables

Text Variables

One-Hot Encoding

Missing Values Imputed

Standardize

Matrix of word-grams occurrences

Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

Prediction

Feature Impact

Data slice All data

Compute method SHAP

Sort by Feature Impact • Decr...

Use quick-compute

Search

Actions

number_inpatient

number_diagnoses

medical_specialty

admission_type_id

discharge_disposition_id

admission_source_id

diag_3

diag_1

age

diag_2

num_lab_procedures

number_emergency

diabetesMed

race

insulin

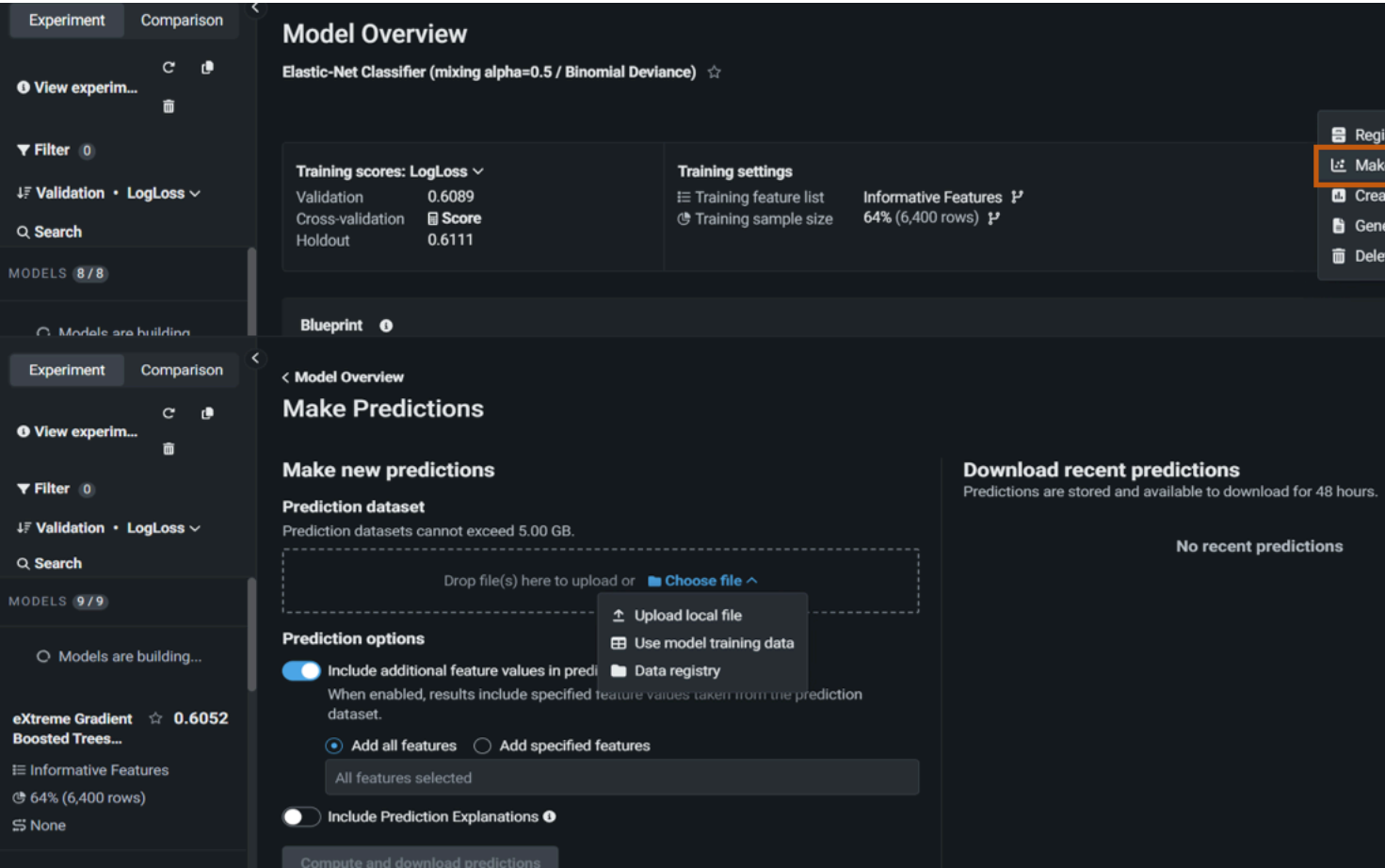
payer_code

change

diag_2_desc

dioplitazone

Step 21: If you have test or unseen data, you can also make predictions by clicking **Make Predictions** under **Model actions**.



Step 22: You can also click **Generate compliance report** and **download compliance report** for your use case.

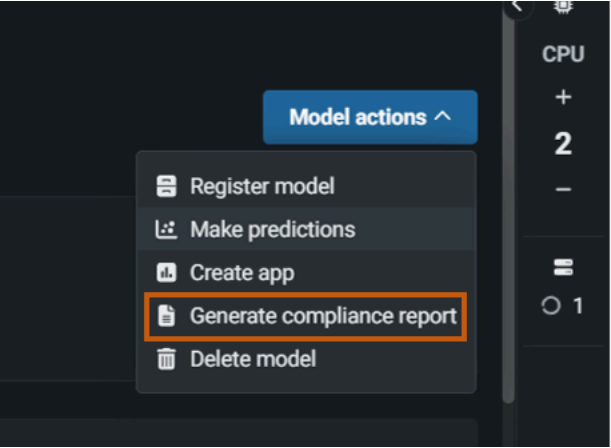


Table of Contents

- 1 How To Use This Document
- 2 DataRobot Model Development Documentation
- 3 Executive Summary and Model Overview
 - 3.1 Model Stakeholders
 - 3.2 Model Development Purpose and Intended Use
 - 3.3 Model Description and Overview
 - 3.4 Overview of Model Results
 - 3.5 Model Interdependencies
- 4 Model Data Overview
 - 4.1 Feature Association
 - 4.2 Data Source Overview and Appropriateness
 - 4.3 Input Data Extraction, Preparation, and Quality & Completeness

Conclusion

In this lab, you have signed up in DataRobot, added a data set in a use case, and worked on data modelling.

Author(s)

[Dr. Pooja](#)



Skills Network